

ENISOLAR

Drone Mission Planning Application

Technical & User Documentation

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Platform: Ionic / Angular (Android & iOS)

Backend: PHP / MySQL (PDO)

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1. Introduction

ENISOLAR is a professional drone mission planning mobile application built for aerial survey and solar panel inspection workflows. The application enables operators to plan, configure, execute, and manage autonomous drone flight missions directly from an Android or iOS device.

The platform combines an interactive satellite map interface with intelligent flight path generation, providing operators with real-time mission statistics and the ability to export waypoint data for direct drone uplink via Bluetooth.

1.1 Purpose

This document provides comprehensive technical and user-facing documentation for the ENISOLAR application. It covers system architecture, feature descriptions, API reference, database schema, and operational guidelines for developers, testers, and end users.

1.2 Scope

The documentation covers all application modules including authentication, mission planning, flight path computation, data persistence, profile management, and drone communication.

1.3 Target Audience

- Drone operators and field technicians using the app in the field
- Mobile developers extending or maintaining the Ionic/Angular frontend
- Backend developers maintaining the PHP/MySQL API layer
- DevOps engineers managing the hosting environment

1.4 Key Capabilities

Capability	Description
User Authentication	Secure registration and login with hashed passwords
Satellite Map Planning	Interactive Google Maps-based polygon drawing tool
Auto Flight Path Generation	Lawnmower-pattern waypoint computation with configurable angle
Real-time Mission Stats	Live calculation of area (ha), flight time, waypoints, and GSD (cm)
Multi-parameter Settings	Altitude, speed, grid angle, front/side overlap, camera model

Capability	Description
Mission Management	Create, save, list, search, and delete missions with cloud sync
CSV Export	Generated flight path exported as order/latitude/longitude CSV
Drone Uplink	Bluetooth BLE connection for direct waypoint upload to drone
Profile Management	Account details and in-app password change functionality

2. System Architecture

ENISOLAR follows a client-server architecture. The mobile client is built with Ionic 6 and Angular 15+, consuming a RESTful PHP backend hosted on a shared hosting environment with a MySQL database.

2.1 Technology Stack

Layer	Technology	Role
Mobile Frontend	Ionic 6 + Angular 15	UI, map interaction, flight path computation
Map Provider	Google Maps SDK	Satellite imagery, polygon drawing, waypoint display
Backend API	PHP 8 (PDO)	REST endpoints, authentication, data validation
Database	MySQL 5.7+	Persistent storage for users and missions
Hosting	AwardSpace (fdb1034.awardspace.net)	Shared PHP/MySQL web hosting
Drone Communication	Bluetooth BLE	Waypoint upload to drone hardware

2.2 Application Layers

2.2.1 Presentation Layer

The Ionic/Angular frontend handles all user interactions. It renders the satellite map, processes touch input for polygon drawing, computes flight statistics in real time, and manages local application state.

2.2.2 Business Logic Layer

Flight path generation logic runs entirely on the client side. The algorithm clips a lawnmower scan pattern to the user-drawn polygon, accounts for the configurable grid angle, and computes inter-waypoint distances. GSD (Ground Sampling Distance) is calculated from altitude and the selected camera specifications.

2.2.3 Data Layer

All mission and user data is persisted to a remote MySQL database via PHP REST endpoints. Missions are identified by UUID strings generated on the client, enabling offline-safe creation before synchronization.

2.3 Data Flow Diagram

The following table summarizes the primary data flows through the system:

Action	Client	Direction	Server
Register	POST full_name, email, password	→	Hash password, INSERT users
Login	POST email, password	→ ←	Verify hash, return user_id
Save Mission	POST mission JSON object	→	UPSERT missions table
Load Missions	GET user_id	→ ←	SELECT missions WHERE user_id
Delete Mission	DELETE id + user_id	→	DELETE WHERE id AND user_id
Change Password	POST user_id, current, new	→	Verify + update hash

3. User Interface & Screen Reference

ENISOLAR presents a dark-themed, professional UI using a navy/black color palette with green accent colors. All interactive elements follow Ionic component conventions optimized for mobile touch interaction.

3.1 Splash Screen

The splash screen displays the ENISOLAR radar-style logo — a green circuit board circular icon on a dark background — while the application initializes and checks for an existing authentication session.

3.2 Login Screen

The login screen presents a centered card on a dark gradient background with the ENISOLAR wordmark at the top.

Fields

- Email address (text input with envelope icon)
- Password (secure text input with lock icon)

Actions

- LOGIN button — authenticates against the PHP backend
- Forgot Password link (visible below the card)
- Create Account button — navigates to the registration screen

Error States

- Email not found — displays inline error message
- Wrong password — displays inline error message
- Network error — displays server error notification

3.3 Registration Screen

The Create Account screen mirrors the login layout with expanded form fields.

Fields

- Full Name
- Email Address
- Password (minimum 8 characters enforced)
- Confirm Password

Actions

- REGISTER NOW — submits registration to the backend
- Sign In link — returns to the login screen

3.4 Map Planner Screen

The Map Planner is the core feature of ENISOLAR. It is a full-screen satellite map view that allows operators to define survey zones and configure flight parameters.

Map Area

Occupies the upper portion of the screen (approximately 55% of viewport height in portrait, full screen in landscape). Uses Google Maps satellite imagery. Tap on the map to add polygon vertices. The survey zone is rendered as a green outline with draggable white vertex handles.

Mission Stats Bar

A dark translucent pill displayed over the map shows four live metrics updated whenever the polygon or settings change:

Icon	Metric	Unit	Description
Arrow	Area	ha	Calculated area of the drawn polygon in hectares
Clock	Flight Time	min s	Estimated mission duration at configured speed
Waypoint	Waypoints	wp	Total number of generated flight waypoints
Eye	GSD	cm	Ground Sampling Distance at configured altitude and camera

Bottom Toolbar

Contains four action buttons:

- Settings (sliders icon) — toggles the Flight Settings bottom sheet
- Undo (curved arrow) — removes the last-added polygon vertex

- Clear (trash icon) — deletes all polygon points and resets the mission
- Done (green checkmark) — finalizes the polygon and generates the flight path

3.5 Flight Settings Panel

A bottom sheet that slides up from below, showing the satellite map in the upper portion and the settings form below. Contains the following controls:

Setting	Type	Range / Options	Default
Mission Name	Text Input	Free text	Enter Mission Name...
Altitude (ALT)	Slider	20m — 200m	60m
Speed	Slider	3 m/s — 20 m/s	10 m/s
Grid Angle	Slider	0° — 90°	0°
Overlap Front	Slider	60% — 95%	80%
Overlap Side	Slider	50% — 90%	70%
Camera Model	Selector	DJI Mini 3 Pro, etc.	DJI Mini 3 Pro

3.6 Inspection / Home Screen

The Inspection screen (accessed via the side menu Home link or after login) displays the list of all saved missions for the authenticated user.

Search Bar

A dark rounded search input allows real-time filtering of missions by name.

Mission Cards

Each mission is presented as a card containing:

- Status badge (READY / DRAFT / etc.) — colored green for ready
- Mission name (bold, white)
- Creation date
- Area in hectares and waypoint count (left column)
- Estimated flight time and GSD in cm (right column)
- Delete button (red trash icon, top-right of card)

Floating Action Button

A purple/dark circular + button in the bottom-right corner navigates to the Map Planner to create a new mission.

3.7 Mission Data Screen

Accessed by tapping a mission card. Displays detailed mission information including the generated CSV flight path and the drone uplink interface.

CSV Flight Path Preview

Renders the generated waypoints in a monospace code block with columns: order, latitude, longitude. A copy icon allows copying the full CSV to clipboard.

Drone Uplink Section

Displays connection status (OFFLINE / CONNECTED) and provides two action buttons:

- CONNECT — initiates a Bluetooth BLE scan to discover and pair with drone hardware
- UPLOAD — transmits the flight path CSV to the connected drone

3.8 Side Menu

Accessible via the hamburger icon. Contains navigation links:

- My Profile — opens profile and password management
- Home — returns to the Inspection screen
- Map Planner — opens a new mission planner
- Logout — clears session and returns to login

3.9 My Profile Screen

Displays the authenticated user's account details and provides a password change form.

Account Details Section

- Username (full name from registration)
- Email address
- Member Since date

Security Section

- Current Password field
- New Password field (min. 8 characters)
- Confirm New Password field
- UPDATE PASSWORD button
- Logout button (top-right corner, red door icon)

4. Flight Path Generation Algorithm

ENISOLAR computes a bounding-box lawnmower scan pattern clipped to the user-defined polygon. The algorithm executes entirely on the client device with no server round-trip.

4.1 Algorithm Steps

1. The user draws a polygon by tapping vertices on the satellite map.
2. On pressing Done, the algorithm computes the axis-aligned bounding box of all vertices.
3. The grid angle setting rotates the scan direction relative to North.
4. Scan line spacing is derived from the camera's sensor width, altitude, and configured side overlap percentage.
5. Horizontal scan lines are generated at the computed spacing across the bounding box height.
6. Each scan line is clipped to the polygon boundary using a ray-casting intersection test.
7. Valid line segment endpoints become flight waypoints, traversed in an alternating left-right serpentine pattern.
8. Total distance is the sum of all inter-waypoint distances computed using the Haversine formula.
9. Estimated flight time = total distance / configured speed, expressed in minutes and seconds.

4.2 GSD Calculation

Ground Sampling Distance (GSD) in centimeters is calculated as:

$$\text{GSD (cm)} = (\text{Altitude} \times \text{Sensor Width}) / (\text{Focal Length} \times \text{Image Width})$$

Camera sensor parameters (sensor width mm, focal length mm, image width px) are stored per camera model within the application. The DJI Mini 3 Pro is the default supported camera.

4.3 Mission Statistics

Statistic	Formula / Source
Area (ha)	Shoelace formula applied to GPS polygon coordinates, converted to hectares
Waypoint Count	Number of clipped scan line endpoints

Statistic	Formula / Source
Total Distance	Sum of Haversine distances between consecutive waypoints
Flight Time	Total Distance / Configured Speed (rounded to seconds)
GSD	Camera model parameters × altitude (see 4.2)

5. Backend API Reference

The ENISOLAR backend exposes a set of PHP REST endpoints hosted at the configured server domain. All endpoints accept and return JSON. CORS headers are set to allow all origins (wildcard) for development purposes.

5.1 Base URL

```
|https://fdb1034.awardspace.net/{endpoint}.php
```

5.2 Authentication Endpoints

POST /register.php

Registers a new user account.

Field	Type	Required	Description
full_name	string	Yes	User's display name
email	string	Yes	Unique email address
password	string	Yes	Plain text password (hashed server-side with PASSWORD_DEFAULT)

Response (success):

```
{ "success": true, "message": "User created" }
```

POST /login.php

Authenticates an existing user. Returns user_id on success.

Field	Type	Required	Description
email	string	Yes	Registered email address
password	string	Yes	Plain text password for verification

Response (success):

```
{ "success": true, "user_id": 42, "message": "Login successful" }
```

Failure responses include: Email not found, Wrong password.

POST /change_password.php

Updates the password for an authenticated user after verifying the current password.

Field	Type	Required	Description
user_id	integer	Yes	Authenticated user identifier
current_password	string	Yes	User's current plain text password
new_password	string	Yes	Desired new plain text password

5.3 Profile Endpoint

GET /get_profile.php

Retrieves account details for the specified user.

Parameter	Type	Required	Description
user_id	integer	Yes	URL query parameter

Response:

```
{ "full_name": "nacer", "email": "user@example.com", "created_at": "2026-02-17" }
```

5.4 Mission Endpoints

POST /save_mission.php

Creates or updates a mission record using UPSERT logic. Supports both new missions and edits to existing ones via the id field.

Field	Type	Description
id	string (UUID)	Client-generated unique mission identifier
user_id	integer	Owner user ID — required for security
name	string	Mission display name

Field	Type	Description
status	string	Mission status (Draft, Ready, etc.)
date	datetime	Creation/modification timestamp
estimatedTime	number	Computed flight duration in seconds
totalDistance	number	Total flight path distance in meters
areaSize	number	Survey area in hectares
waypointCount	integer	Number of waypoints in flight path
config.altitude	number	Flight altitude in meters
config.speed	number	Flight speed in m/s
config.gridAngle	number	Scan grid angle in degrees
config.overlapFront	number	Front overlap percentage
config.overlapSide	number	Side overlap percentage
config.cameraModel	string	Selected drone camera model name
polygonPoints	JSON array	Array of {lat, lng} objects defining the survey zone
flightPath	JSON array	Array of {lat, lng} waypoint objects

GET /get_missions.php

Returns all missions for a user, or a single mission when mission_id is provided.

Parameter	Required	Description
user_id	Yes	Filter missions by owner
mission_id	No	If provided, returns a single mission object instead of array

GET /get_mission_details.php

Alternative endpoint returning a fully expanded mission object including all config fields, polygon points, and flight path. Optimized for the Mission Data screen.

DELETE /delete_mission.php

Deletes a mission. Accepts id and user_id via URL parameters, JSON body, or DELETE method body. The user_id constraint ensures users can only delete their own missions.

Parameter	Required	Description
id	Yes	Mission UUID to delete
user_id	Yes	Security constraint — must match mission owner

Returns HTTP 403 if the mission is not found or does not belong to the requesting user.

6. Database Schema

6.1 users Table

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
full_name	VARCHAR(255)	NOT NULL	User's display name
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email address
password	VARCHAR(255)	NOT NULL	bcrypt hashed password
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation time

6.2 missions Table

Column	Type	Description
id	VARCHAR(255) PRIMARY KEY	Client-generated UUID
user_id	INT NOT NULL	Foreign key to users.id
name	VARCHAR(255)	Mission display name
status	VARCHAR(50)	Mission state (Draft, Ready, etc.)
created_at	DATETIME	Mission creation timestamp
estimated_time	FLOAT	Estimated flight duration (seconds)
total_distance	FLOAT	Total flight path distance (meters)
area_size	FLOAT	Survey polygon area (hectares)
waypoint_count	INT	Number of flight waypoints
altitude	FLOAT	Configured flight altitude (meters)
speed	FLOAT	Configured flight speed (m/s)
grid_angle	FLOAT	Scan grid angle (degrees)
overlap_front	FLOAT	Front image overlap (%)
overlap_side	FLOAT	Side image overlap (%)
camera_model	VARCHAR(100)	Selected camera model name
polygon_points	JSON / TEXT	JSON array of survey zone vertices
flight_path	JSON / TEXT	JSON array of flight waypoints

7. Security Considerations

7.1 Implemented Security Measures

- Password hashing using PHP PASSWORD_DEFAULT (bcrypt with automatic salting)
- Parameterized queries (PDO prepared statements) throughout all SQL operations, preventing SQL injection
- Mission delete endpoint validates user_id ownership before deleting, preventing unauthorized data deletion
- User_id required and validated on all mission write/delete operations

7.2 Known Limitations & Recommendations

Issue	Risk	Recommendation
No JWT or session tokens	High	Implement token-based auth (JWT). Currently user_id is passed in plaintext by the client.
CORS set to wildcard (*)	Medium	Restrict Access-Control-Allow-Origin to known app origins in production
Database credentials in source	High	Move db_connect.php credentials to environment variables or server config files outside web root
No rate limiting	Medium	Add login attempt rate limiting to prevent brute force attacks
HTTPS not verified	High	Ensure all API calls use HTTPS. AwardSpace supports SSL certificates.
No input length validation	Low	Add server-side validation for field lengths and data types

8. Installation & Deployment

8.1 Backend Deployment

10. Upload all PHP files to the web root of the hosting account (fdb1034.awardspace.net)
11. Ensure db_connect.php has the correct database credentials
12. Create the MySQL database and run the table creation SQL (users and missions tables)
13. Verify PHP version is 7.4+ with PDO and PDO_MySQL extensions enabled
14. Confirm all files are accessible over HTTPS

8.2 Mobile App Build

15. Clone the Ionic/Angular source repository
16. Run npm install to restore all dependencies
17. Update the API base URL constant to match the production server domain
18. Run ionic build --prod for a production build
19. Run ionic capacitor add android (or ios) to add native platforms
20. Use Android Studio or Xcode to build and sign the application package

8.3 Environment Configuration

Variable	Location	Description
\$host	db_connect.php	MySQL server hostname
\$db_name	db_connect.php	Database name
\$username	db_connect.php	Database username
\$password	db_connect.php	Database password
API_BASE_URL	Angular environment.ts	Base URL for all HTTP requests

9. Known Issues & Roadmap

9.1 Known Issues

ID	Severity	Description
KI-001	Medium	GSD value shows blank (cm) on mission cards in the list view when camera_model is not saved
KI-002	Low	Map initializes at global view on first load before centering to user location
KI-003	Medium	Drone uplink remains OFFLINE placeholder; full BLE implementation to be verified on hardware
KI-004	Low	logout link in side menu uses lowercase — inconsistent with other menu items

9.2 Planned Features

- JWT-based session authentication for stateless and secure API access
- Offline mode with local mission queue and background sync when connectivity is restored
- Support for additional drone camera models and custom sensor parameters
- KML/GeoJSON export formats in addition to CSV
- Mission status lifecycle management (Draft → Ready → Completed → Archived)
- Live telemetry visualization during active drone missions
- Multi-mission batch planning for large survey areas
- Team/organization accounts with shared mission libraries

10. Glossary

Term	Definition
GSD	Ground Sampling Distance — the real-world size (cm) represented by a single image pixel at the configured altitude
Waypoint	A geographic coordinate point (lat/lon) that the drone navigates to in sequence
Lawnmower Pattern	A serpentine back-and-forth scan pattern used in aerial survey to achieve uniform coverage
Overlap (Front)	Percentage of image overlap between consecutive images along the flight direction
Overlap (Side)	Percentage of image overlap between adjacent flight lines
UUID	Universally Unique Identifier — used as the mission primary key to allow client-side generation
BLE	Bluetooth Low Energy — the wireless protocol used to communicate with drone hardware
UPSERT	Database operation that inserts a new record or updates it if the primary key already exists
PDO	PHP Data Objects — PHP database abstraction layer that enables parameterized queries
CORS	Cross-Origin Resource Sharing — HTTP headers that control which origins may access the API
bcrypt	Adaptive password hashing function used by PHP PASSWORD_DEFAULT

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